

Improving the efficiency of sentiment analysis system based on integration with Robotic Process Automation and Knowledge-based System

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Abstract. In the digital landscape, sentiment analysis plays a crucial role in processing large volumes of data for timely decision-making. This research introduces a novel integration of Robotic Process Automation (RPA) and Knowledge-based Systems (KBS) for sentiment analysis, specifically designed to address limitations in existing systems. Unlike traditional methods, which rely heavily on predefined rules or static keyword sets, this approach uses the PhoBERT model for advanced Vietnamese text processing and uses cosine similarity to classify data more accurately. By automating the collection and segmentation of learner feedback on social networks, the system achieves approximately 80% accuracy in classifying sentiment when deployed at several universities in Vietnam. The integration of RPA enhances the efficiency of the system by reducing manual intervention, while KBS ensures real-time updates to the knowledge base, overcoming the dependency on external databases. This new integration provides a scalable solution, improves data processing speed, and increases adaptability compared to traditional sentiment analysis models, helping higher educational institutions understand and engage with learners. These findings can be applied and further extended to a wide range of practical domains.

Keywords: Robotic Process Automation, RPA, Knowledge-Based System, KBS, Sentiment Analysis System.

1 Introduction

In the context of rapid digital transformation, universities and organizations face enormous volumes of data from social networks, emails, and user feedback. Exploring sentiment analysis from data using traditional methods faces many challenges. Previous studies have explored various methods to improve sentiment analysis, such as automatically collecting feedback [9], then using deep learning models [4] or hybrid models like IChOA-CNN-LSTM for multilingual emotion recognition [2]. However, these methods still face limitations, particularly when

dealing with large, unstructured datasets and the need for real-time data processing. Furthermore, many existing solutions are unable to adapt quickly to changing sentiment or update keyword sets automatically, leaving them vulnerable to outdated data or inaccurate classifications.

To address these challenges, the study proposes an innovative integration of RPA with KBS. Unlike traditional models, the integration of RPA and KBS will help automate data collection, update knowledge bases in real time, and accurately classify data using advanced techniques such as the PhoBERT model for Vietnamese text processing. This solution not only improves efficiency by automating repetitive tasks but also ensures flexibility and scalability, enabling the system to handle large volumes of data with greater accuracy. This study addresses two core questions as follows:

- **RQ1:** How can RPA and KBS be effectively integrated into data segmentation?
- **RQ2:** How can the system efficiently manage a large keyword repository and classify data into departments without continuous database access?

RQ1 focuses on how to effectively integrate RPA and KBS into data segmentation, aiming to optimize and automate the classification process. Through automated data retrieval using RPA and the development of a dynamic knowledge base via API integration, the system uses semantic mapping along with constructing composite vectors with the PhoBERT model to accurately group unstructured data into corresponding departments. Meanwhile, **RQ2** addresses the use of a dynamic knowledge base in memory, enabling the system to eliminate the need for continuous database access. By using keywords through API integration, the system reduces dependency on external databases and ensures fast and efficient data retrieval. The data segmentation mechanism, based on thresholds calculated using cosine similarity, optimizes classification performance and allows for real-time updates across the system.

Solving the above two research questions is expected to help build a successful sentiment analysis system with scalability and high accuracy suitable for higher education institutions (HEIs), while paving the way for wide application in many other fields such as business, management, customer service, etc.

2 Methodology

2.1 Automated using RPA

RPA is a powerful and efficient solution for automating repetitive tasks in sentiment analysis systems [8]. RPA provides an integrated, low-code environment that reduces development time, minimizes human error, and enables seamless interaction with legacy systems via GUI automation capabilities often limited in traditional scripting approaches [1]. For this study, the automated activities covered included: (1) *collecting data from social networks* and (2) *responding via email to relevant departments*.

Data collection: To be able to collect data, using an RPA solution (*such as UiPath,...*) to collect data shared by learners is considered effective compared to the effort of manually accessing and monitoring learners' views from social networks proactively initiated by HEIs. RPA helps to automatically extract data (*such as posts, comments, etc.*) to provide valuable insights into the development of learners' thinking. Unlike manually collected data sets which are limited in size, incomplete and laborious.

Send warning emails: A proposed component is an automated email distribution process, ensuring timely and accurate distribution of processed information to relevant departments. This component retrieves data from the existing database of student posts and comments in real time. Next, this data is merged with sentiment analysis results to filter out negative content (*based on Artificial Intelligence*) to be sent. At the same time, it is merged with the analysis results of which department in the higher education institution that content belongs to (*based on KBS*). The RPA solution will automatically send warning emails of negative student feedback to relevant departments in the higher education institution. This automation process not only speeds up the flow of information but also minimizes manual effort and reduces the risk of human error.

Using RPA helps improve the performance of sentiment analysis systems. Automated data collection is the basis for performing real-time sentiment trend analysis. And with the analysis results, automatically identifying and sending the results to the relevant departments in a timely manner via email allows for timely intervention, helping HEIs increase engagement with learners.

2.2 Using the Knowledge Base to segment departments

This activity was tested after automatically collecting more than 10,000 articles shared by learners on social networks initiated and managed by a Vietnamese university through RPA. The data was pre-processed to remove noise (*e.g., irrelevant symbols, stop words*) and normalize the text to ensure consistent analysis. From there, calculations were performed to segment the data into corresponding departments.

Methods for utilizing the Knowledge Base: The system retrieves keyword and department information via APIs, creating a dynamic knowledge base that allows real-time updates without direct interaction with the local database. Developed using Python with support from libraries like requests (*for API retrieval*), json (*for data formatting*), and uuid (*for generating unique identifiers*), the obtained data is organized into a dictionary stored in memory from the out-set. This method ensures fast querying and efficient resource utilization. Any changes to keywords or department structures trigger an automatic synchronization process, ensuring consistency between the external API data and the internal processing pipeline. Integration with the databases via APIs increases system flexibility, especially in distributed environments with data updated from multiple sources.

Cosine Similarity calculation: To perform automatic classification, the system integrates the PhoBERT language model, a deep learning model trained

on Vietnamese text data [7] and uses the cosine similarity method [3]. In practice, the system focuses on similarity values between 0 and 1 to identify the most relevant departments.

Formula for Cosine Similarity:

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

$\mathbf{A} \cdot \mathbf{B}$: is the dot product of two vectors (*sum of element-wise multiplications*).

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=0}^n A_i \times B_i$$

$\|\mathbf{A}\|, \|\mathbf{B}\|$: are the magnitudes (*norms*) of vectors A and B.

$$\|\mathbf{A}\| = \sum_{k=0}^n A_k$$

$$\|\mathbf{B}\| = \sum_{k=0}^n B_k$$

The use of cosine similarity in recommendation systems or natural language processing tasks for input data has clear advantages [3]. First, cosine similarity helps measure the degree of similarity between sentences or texts by calculating the angle between the vector representations of the sentences, without being influenced by their length. This makes it a powerful tool when comparing sentences of different lengths, as it focuses solely on the direction of the vectors rather than their absolute length. Secondly, cosine similarity is very effective when applied to sentence embedding models, especially with models like BERT in this study, which helps capture the semantic characteristics of sentences without worrying about the difference in sentence lengths. The use of cosine similarity also helps optimize accuracy in tasks such as measuring semantic textual similarity, text clustering, and text classification. Finally, when combined with extraction techniques from Transformer models, cosine similarity not only improves the performance of embedding models but also makes the system flexible and applicable to various tasks in natural language processing.

After the calculation, the program performs an important step of determining thresholds to assess the degree of similarity between the input text vector and the average vector of departments (*department embeddings*), calculated using cosine similarity. These thresholds not only help differentiate the confidence levels of the classification results but also provide a scientific basis for balancing precision and recall in the system as shown in Table 1.

Through carefully selected classification thresholds, the system can automatically classify texts with high reliability, while also providing a foundation for applying additional processing measures for cases with ambiguous similarity levels.

Threshold (%)	Accuracy	Recall	Precision	F1-Score
30	0.202	1.000	0.202	0.336
35	0.202	1.000	0.202	0.336
40	0.253	1.000	0.213	0.351
45	0.313	1.000	0.227	0.370
50	0.465	1.000	0.274	0.430
55	0.677	1.000	0.436	0.607
60	0.859	0.950	0.594	0.731
65	0.859	0.600	0.667	0.632
70	0.838	0.300	0.750	0.429

Table 1: Model performance at various cosine similarity thresholds

3 Results

3.1 Propose a process for performing collect and divide data across departments based on RPA and KBS

After identifying the problem and considering an appropriate implementation solution to collect and categorize learners' opinion data related to departments within a university, a proposed process for data collection, classification and processing is presented in Figure 1.

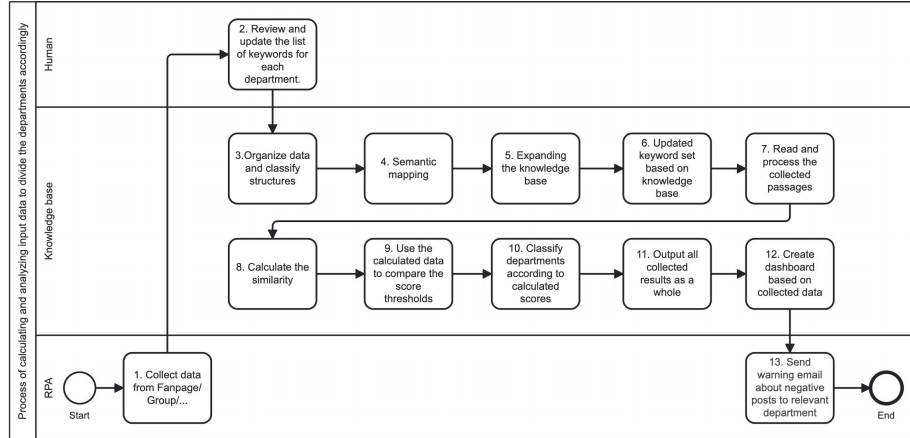


Fig. 1: Workflow for analyzing input data and assigning it to relevant departments

The process comprises 13 main steps supported by components such as humans, a knowledge base, and software robots. The input data is the views of learners on social networks, which are automatically collected by the software robot. Humans review and update the list of keywords for each department, then input these keywords into the knowledge base to allocate the analyzed data to

the respective sections. Meanwhile, the knowledge base stores the set of keywords for each department, thereby supporting the computation of similarity between the input data to assign them to the appropriate department. Finally, the software robot will automatically send alert emails related to segmented data to relevant departments.

3.2 Applying UiPath platform to automatically collect data and send warning emails

Two software robots built on the UiPath platform to automate data collection and send email alerts regarding learners' opinions are shown in Figure 2.

Data collection: The robot built on the UiPath platform will collect data including student posts, comments and related data. The output from the data collection is integrated directly via API, ensuring that all learner-generated content is easily accessible for downstream processing.

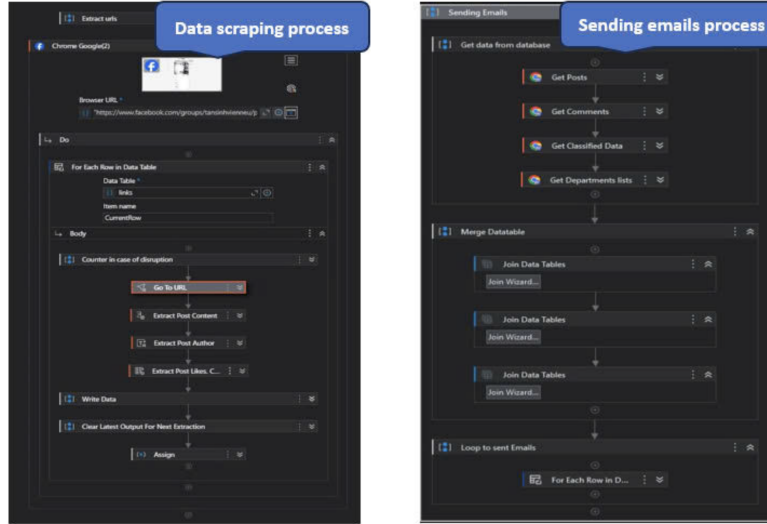


Fig. 2: Design of Robots for automated data collection and warning email delivery

Send warning emails: After identifying the negative content and the corresponding relevant departments, the process will generate alert email content to the relevant departments. This application ensures that relevant departments will receive timely updates on negative content related to their department mentioned by learners. As a result, relevant parties can recognize early and make timely interventions to avoid unfortunate communication incidents.

3.3 Develop a system in analyzing and dividing into departments with a set of knowledge bases

To have a complete system, it will be divided into 2 main functions: creating a knowledge base and dividing data into corresponding departments.

Knowledge Base: At this stage, each unit provides a carefully selected set of keywords that accurately reflect its area of expertise. These keywords are stored through an API, allowing real-time retrieval of keyword and unit information, thereby building a dynamic knowledge base in memory. This improves query speed and reduces system load.

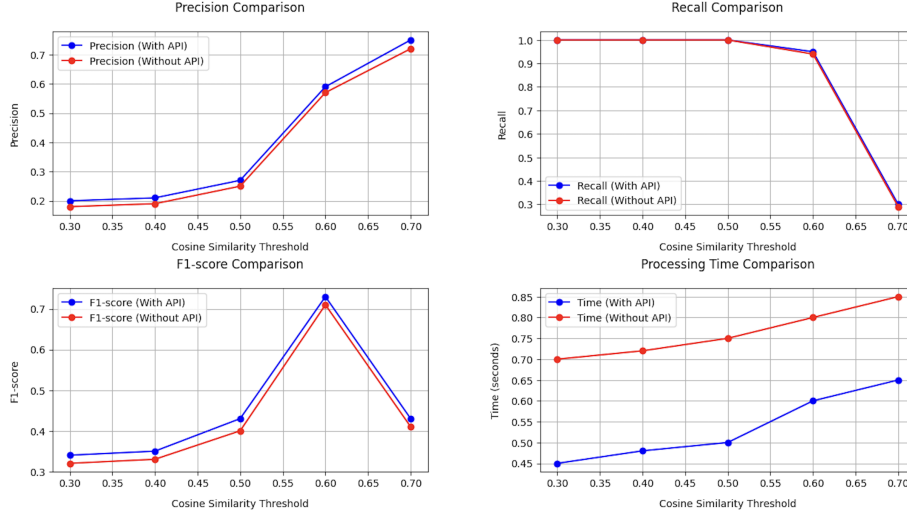


Fig. 3: Comparison of system performance with and without API integration

The analysis results presented in Figure 3 show that the system integrated with an API significantly outperforms the one without an API in both processing time and classification performance. Specifically, the API-enabled system processes queries considerably faster, exhibiting a lower average response time. This improvement is attributed to the API integration, which reduces dependency on external databases and enhances in-memory data retrieval efficiency.

In terms of classification performance, the API-based system consistently achieves higher values in precision, recall, and F1-score across most cosine similarity thresholds. Notably, at the 0.6 threshold, the F1-score peaks at 0.73, indicating an optimal balance between precision and recall. In contrast, the system without an API, while maintaining relatively high recall at lower thresholds, experiences a significant drop in precision and F1-score at higher thresholds. This suggests limitations in its ability to accurately handle and classify data. These findings indicate that API integration not only accelerates processing but also substantially enhances the effectiveness of text classification.

Data classification: To evaluate the proposed approach, an experiment was carried out to automatically collect more than 10,000 posts and comments shared by learners on social networks administered by a Vietnamese university. A total of 100 samples were selected for training and testing purposes. This selection aimed to reduce noise and improve classification accuracy, despite the limited sample size. The strength of the dataset lies in its authenticity, as all data

were derived from genuine student responses, thereby reflecting the real-world context and issues they are encountering. Subsequently, cosine similarity was employed to compare the computed samples against manually classified texts through direct reading.

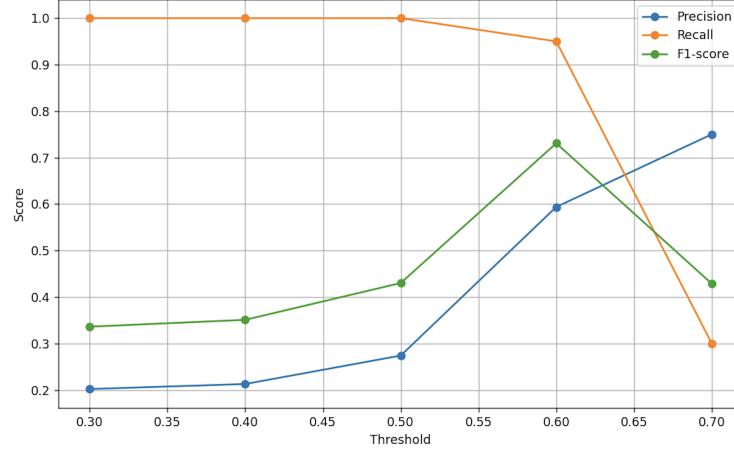


Fig. 4: Chart of performance indicators across cosine similarity thresholds

Based on the experimental results illustrated in Figure 4, performance metrics such as precision, recall, and F1 score to evaluate the accuracy of cosine similarity were used. The results show that with a threshold of 0.60, precision reaches 59%, recall remains at 95%, and the F1-score achieves its optimal value of 0.73, corresponding to a system accuracy of $73\% \pm 2\%$ (*with a confidence interval 95%*). Detailed statistical analysis, including standard deviation testing and t-test validation, indicates that the 0.60 threshold outperforms the basic keyword matching model ($p < 0.05$). This confirms that the 0.60 threshold is the optimal choice, achieving an ideal balance between precision and recall, while also validating the reliability of the classification results. Furthermore, the modular design of the system is highly flexible, allowing for its application in domains beyond education. For instance, in the customer service sector, this method could be adapted for sentiment analysis to prioritize urgent complaints by classifying responses into appropriate departments.

Thus, it can be seen that the integration of a robust knowledge base, advanced text feature extraction with PhoBERT, and a comprehensive experimental process has demonstrated that the proposed system achieves high performance in the educational context, while also showing significant potential for application in various other fields.

4 Discussion

The first challenge in sentiment analysis is to ensure the legality of the input data source for analysis and processing, which must comply with the regulations

of each country. Ideally, the data should be derived from public pages and groups initiated and managed by HEIs. As a result, fully capturing all the nuances and perspectives of real learners becomes challenging. Furthermore, social networks frequently alter their data-sharing policies, making it difficult to develop a solution that guarantees long-term stability.

Additionally, the cost of implementing automation systems is an important consideration. Advanced RPA solutions like UiPath Document Understanding offer intelligent automation capabilities but require significant licensing and infrastructure costs [6]. These costs could become a major barrier, particularly in educational environments, where resources are often limited.

Moreover, modern methods in text analysis, such as fine-tuning PhoBERT directly on large labeled datasets, have achieved impressive accuracy levels of 85-90%. However, these methods require significant computational resources [5]. To address this, our system uses PhoBERT for feature extraction combined with cosine similarity. This approach strikes a balance between performance and resource requirements, reducing the need for model training, saving computational resources, while still maintaining an accuracy of around 80%, suitable for practical needs. Furthermore, relying on a fixed set of keywords makes it difficult to keep up with the rapidly changing sentiments of students. To overcome this limitation, it is necessary to integrate a self-learning mechanism using machine learning models to automatically update the keyword set and expand multilingual capabilities.

Our RPA solution, combined with KBS not only keeps costs low and is easy to deploy but also provides stable accuracy, making it well-suited for the needs of HEIs under current practical conditions. The potential to expand this solution to other domains further promises to enhance efficiency and provide a competitive advantage.

5 Conclusion and Future work

This study demonstrates the integration of RPA and KBS for sentiment analysis, which significantly improves the process of collecting, classifying, and processing learner sentiment data. The automated workflow not only reduces manual labor but also improves the speed and efficiency of decision-making processes. The system achieves approximately 80% accuracy in classifying learner feedback, as evidenced by experiments conducted at a Vietnamese university, with a well-balanced precision-recall performance at the 0.60 threshold.

Despite the positive outcomes, the system still faces several challenges that need to be addressed, such as ensuring stable data collection, automatic keyword suggestion, and multilingual processing. To gain a more comprehensive understanding of learners' perspectives, it is essential to diversify the sources of legitimate data beyond social networks. Additionally, the development of a self-learning mechanism, based on machine learning models such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and reinforcement learning, should be explored. This mechanism could automatically update

the keyword set, allowing the system to maintain high accuracy without human intervention. Furthermore, the system needs to be further applied in various contexts and domains (*e.g., business, management, customer service, etc.*) to make it more adaptable to multicultural and multilingual environments.

The integration of RPA and KBS into the sentiment analysis system not only enhances efficiency but also highlights the technological advancements of the solution. By automating repetitive tasks, RPA improves productivity, increases accuracy, and reduces human intervention. Meanwhile, KBS provides a knowledge base that enables the system to classify and distribute data automatically and accurately to stakeholders. This combination opens the door for large-scale deployment, benefiting HEIs in particular and a wide range of sectors in the economy in general.

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